

Hardware Components & System Assembly- assament

Section- A

SCENARIO -1 : You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Q- You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

A- Component selection should start with the CPU, since it drives every other choice. For an office workstation running productivity software (Word, Excel, browser, email), a mid-range CPU such as an Intel Core i3/i5 or AMD Ryzen 3/5 is sufficient — there is no need to over-spend on a high core-count chip. Once the CPU is chosen, the motherboard is selected to match it:

- **Socket match:** the motherboard's CPU socket (e.g. LGA1700 for recent Intel chips, AM5/AM4 for Ryzen) must exactly match the CPU's socket type.
- **Chipset & form factor:** the chipset must support the CPU generation, and the board's form factor (ATX/mATX/ITX) must fit the intended case.
- **RAM type/speed support:** the board's specification sheet lists the supported memory type (DDR4 or DDR5), maximum speed, and number of slots — the RAM purchased must match this type and ideally appear on the manufacturer's Qualified Vendor List (QVL).
- **Storage interfaces:** confirm the board has the required number of M.2 (NVMe) slots and/or SATA ports for the chosen storage device(s).

To confirm everything will actually work together before purchase, I would:

(1) check the CPU manufacturer's socket/spec page against the motherboard's official socket listing

(2) check the motherboard's QVL for the specific RAM kit

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- (3) verify the motherboard's BIOS version supports the CPU generation — newer CPUs sometimes need a BIOS update on older boards, which may require a compatible "boot-strap" CPU
- (4) confirm physical clearance for the CPU cooler and RAM height inside the chosen case and
- (5) cross-check the whole parts list in a com

S2 — SCENARIO You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

A- For a workload that is mainly office/productivity applications, a SATA SSD is the most suitable recommendation. Office apps and web browsing are not throughput-intensive, so the extra bandwidth of NVMe brings little day-to-day benefit for this workload, while its added cost is better spent elsewhere. A SATA SSD still gives a large, very noticeable improvement over an HDD in boot time, application launch time, and general responsiveness, and it is far more reliable since it has no moving parts. An entry-level NVMe drive can be recommended if the budget allows and the motherboard has a free M.2 slot, purely for a snappier feel, but it is an optional upgrade rather than a necessity. HDDs are not recommended as the primary/boot drive for this client, though one can still be added as low-cost secondary bulk storage for file archives if needed.

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S3 — You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

A-

Role of the GPU: the GPU offloads rendering and compute-heavy operations — 2D/3D viewport rendering, layer compositing, video preview acceleration — from the CPU. A dedicated GPU with adequate VRAM prevents the lag and dropped frames that occur when a design application has to fall back on integrated graphics or software rendering for large files. The GPU's drivers must also support the graphics API the design software relies on (DirectX/OpenGL/CUDA), otherwise features may be disabled or run in a slower fallback mode.

Role of the cooling system: several hours of sustained load keeps the CPU and GPU near their maximum operating temperature for long periods, which is a very different thermal condition to short bursts of use. If cooling is inadequate, the components will thermally throttle — automatically reducing their clock speed to stay within a safe temperature — causing the exact slowdowns and stutter the user is trying to avoid. Beyond throttling, sustained high temperatures increase the risk of instability (crashes, driver resets) and shorten component lifespan. A reliable setup for this scenario needs good case airflow (balanced or slightly positive-pressure intake/exhaust fan arrangement), an adequately sized CPU cooler for the chip's TDP, unobstructed GPU airflow, and periodic dust cleaning, since dust buildup is a common cause of gradually rising temperatures over months of daily use.

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S4 — You are assembling a PC from components sourced from three different vendors.

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

A-

- Socket/physical mismatch: CPU does not fit the motherboard socket at all.
- Unsupported RAM: memory not on the board's QVL, or the wrong DDR generation, causing failure to boot or random crashes/blue-screens under load.
- Insufficient/incompatible PSU: wrong wattage or missing connectors (e.g. no 8-pin EPS, no PCIe GPU power) prevents the system from powering the GPU or even from booting.
- Physical clashes: a GPU or CPU cooler too large for the case, or a motherboard form factor that doesn't fit the case's standoff layout.
- BIOS/firmware gaps: a motherboard shipped before a newer CPU generation existed may not POST until its BIOS is updated.
- Support/warranty friction: when a fault occurs, mixed-vendor builds make it harder to pinpoint which vendor is responsible, slowing down replacement or repair.

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S5 — You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

A-

Choosing the PSU rating: add up the TDP of the CPU and GPU (the two largest consumers) plus a reasonable allowance for storage devices, fans, and other peripherals, then add roughly 20–30% headroom for power-supply efficiency, transient spikes, and future upgrades. I would choose a unit from a reputable manufacturer with an 80 PLUS efficiency rating (Bronze/Gold or better) and confirm it physically has the required connectors: a 24-pin ATX main connector, a 4/8-pin EPS connector for the CPU, PCIe 6+2-pin connector(s) for the GPU, and SATA/Molex connectors for drives and fans.

Safety and connection checks before first power-on:

1 Keep the PSU switched off and unplugged from the wall while making all connections.

2 Use ESD precautions (anti-static wrist strap, or regularly touch a grounded metal surface) before handling any component.

3 Confirm the PSU's voltage selector (115V/230V) is set correctly for the region — an incorrect setting can damage the whole system.

4 Seat and fully click in the 24-pin motherboard connector and the CPU 4/8-pin EPS connector.

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5 Connect GPU PCIe power cables directly from the PSU (avoid daisy-chaining if the GPU needs sustained high power).

6 Check that no loose screws or metal off-cuts are resting on the motherboard, and that only the correct number of standoffs are installed under the board to avoid a short circuit.

7 Route cables away from fan blades and confirm nothing is pinched under a bracket.

S6 —You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

A-

Checks to confirm a successful POST:

- Power LED and case/CPU fans spin up immediately on power-on.
- Monitor displays the motherboard/BIOS splash screen (confirms GPU output and cable/input source are correct).
- No beep codes or on-screen POST error codes are reported.
- BIOS/UEFI correctly identifies the installed CPU model and full RAM capacity (e.g. 16 GB shown, not 8 GB — which would indicate a seating issue).
- The storage device(s) appear in the BIOS boot/storage menu.
- System date/time can be set and holds after a restart (confirms the CMOS battery and RTC are functioning).

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- System proceeds to boot into the installed OS (or shows a "no bootable device" message only because an OS hasn't been installed yet, which is expected at this stage, not a hardware fault).

If POST fails, I would troubleshoot in this order:

- 1 No display / no beep: reseal the RAM (try one stick at a time, in different slots), confirm the GPU is fully seated and its power connector is attached, and check the monitor cable/input source.
- 2 Beep codes present: consult the motherboard manual to decode the beep pattern (patterns typically point to RAM, GPU, or CPU issues).
- 3 Re-check that the 24-pin and CPU 8-pin power connectors are fully seated — a partially seated CPU power connector is a very common cause of a dead board.
- 4 Inspect the CPU socket for bent pins (Intel) or check that the CPU is properly locked down (AMD) if this is a new install.
- 5 Clear the CMOS (jumper or battery pull) in case of a corrupted BIOS setting from a previous configuration.
- 6 Strip the build down to a minimal configuration (CPU + one RAM stick + PSU, outside the case) to isolate which single component is at fault before reassembling.

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Section B

Task 1: Component Identification & Functions

Component:- Motherboard

Function:- Central circuit board connecting and allowing communication between all components

Component:- CPU

Function:- Executes instructions; the primary processing unit of the system

Component:- ram

Function:- Temporary high-speed working memory for actively used data/programs

Component:- Storage device

Function:- Non-volatile long-term storage for the OS, programs, and files

Component:- Cooling system

Function:- Removes heat from the CPU/GPU/case to prevent thermal throttling and damage

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Task 2 — Motherboard Ports & Socket Identification

Major ports/headers typically identified on the board:

- 24-pin ATX main power connector — supplies primary power to the motherboard.
- 4/8-pin CPU (EPS) power connector — dedicated power to the CPU.
- SATA data ports (usually 4–6) — connect SATA SSDs/HDDs and optical drives.
- Front-panel header — connects case power/reset buttons and status LEDs.
- USB headers (internal, for front-panel USB) and rear-panel USB ports (USB-A/USB-C).
- Fan headers (CPU_FAN, SYS_FAN) — power and control case/CPU fans.

Expansion slots and purpose:

- PCIe x16 slot — primary slot for the GPU, providing the highest bandwidth lane group.
 - PCIe x1/x4 slots — for smaller add-in cards (capture cards, extra USB/network cards).
 - M.2 slot(s) — mounts NVMe SSDs directly onto the board using PCIe lanes for high-speed storage (some boards also support M.2 SATA/Wi-Fi cards).
- CPU socket: identify the socket type printed near the CPU mount (e.g. LGA1700 for recent Intel generations, AM5 for current Ryzen). Cross-reference the motherboard manufacturer's product page.

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Task 3 — RAM & Storage Installation

RAM removal/reinstallation procedure:

- 1 Power off the system fully and unplug it from the wall.
- 2 Put on an anti-static wrist strap (or touch a grounded metal case part) before touching any component — ESD precaution.
- 3 Press outward on the retention clips at both ends of the RAM slot until the module releases, then lift it straight out
- . 4 To reinstall: align the notch on the module with the key in the slot, press straight down evenly on both ends until the retention clips click back into the locked position.
- 5 Confirm proper seating visually (module sits flush, clips fully closed) and functionally by booting into BIOS and checking the full, correct capacity is listed.

Storage device removal/reinstallation procedure:

- 1 For a SATA drive: disconnect the SATA data cable and SATA power cable, then unscrew the drive from its bay/bracket.
- 2 For an NVMe drive: unscrew the single retention screw at the end of the M.2 slot, then gently lift the drive out of the slot at its resting angle.
- 3 Reinstall by reversing the steps — reseal the SATA cables firmly and rescrew the drive, or for M.2, insert at the angled position and press down before securing the screw.
- 4 Confirm detection by entering BIOS/UEFI and checking the storage device appears in the boot/drive list with the correct model and capacity

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Task 4 — Full Desktop PC Assembly

- 1 ESD precaution: wear an anti-static strap/mat for the entire build.
- 2 Install the CPU onto the motherboard (outside the case) — align the corner triangle/notch markers, lower gently, close the retention arm without forcing it.
- 3 Apply thermal paste (if not pre-applied on the cooler) and mount the CPU cooler, connecting its fan cable to the CPU_FAN header.
- 4 Install RAM into the motherboard's recommended slots (usually alternating slots for dual-channel) before mounting the board in the case, since it is easier to seat with the board on a flat surface.
- 5 Mount the motherboard into the case on the correct standoffs, and secure with screws.
- 6 Install the storage device(s) (M.2 directly on the board, or SATA into drive bays with cables run to the board).
- 7 Install the PSU into its bay and route cables through the case's cable-management cutouts.
- 8 Connect the 24-pin and CPU 8-pin power connectors, front-panel header, USB headers, and fan headers.
- 9 Install the GPU into the primary PCIe x16 slot, secure it to the case bracket, and connect its PCIe power cable(s) from the PSU.
- 10 Double-check all connections and confirm no loose screws/debris are inside the case before closing it.
- 11 Connect a monitor, keyboard, power cable, and switch on the PSU, then power on the system and confirm it passes POST.

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Section C — Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis — ABC Solutions

1. Three components to check first when a newly assembled PC fails to boot

- RAM: the single most common cause of a no-boot/no-display state after assembly — often just a seating issue, and quick to test by reseating or trying one stick at a time.
- Power connections (PSU to motherboard/CPU/GPU): a missing or partially seated 24-pin or CPU 8-pin connector will prevent the board from powering on at all, so this is checked immediately after RAM.
- GPU seating and power: on builds with a discrete GPU, an unseated card or missing PCIe power cable is a very common cause of "no display" even when the board itself is powering on correctly. These three are checked first because they are the fastest to physically verify, do not require any tools beyond a screwdriver, and together account for the large majority of first-boot failures in freshly assembled systems.

2. Motherboard–CPU compatibility and system functionality

On the system assembled/inspected for this project (see Task 4), the motherboard's socket type was matched to the CPU's socket before assembly, and the motherboard's supported-CPU list was checked to confirm the specific CPU model/generation was supported at the installed BIOS version. If this step is skipped, the system will typically fail to POST entirely — the board may not power the CPU correctly, may show no display at all, or in the worst case the CPU may not even physically fit the socket. Even when the socket physically matches, an out-of-date BIOS on the motherboard can prevent it from recognizing a newer CPU generation,

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which is a compatibility failure that looks identical to a hardware fault (no POST) unless it is specifically checked.

3. Verifying RAM and storage installation via BIOS/POST

RAM is verified by entering the BIOS/UEFI setup and confirming the total capacity shown matches what was physically installed (e.g. two 8 GB sticks should report 16 GB) — a lower reported capacity indicates one module is not seated correctly. The BIOS memory information screen should also confirm the correct number of populated slots and, ideally, the rated speed. Storage is verified by checking the BIOS storage/boot device list: the installed SSD/HDD should appear with its correct model name and capacity. If it does not appear, the cause is almost always a loose SATA/power cable or, for NVMe, an improperly seated M.2 module or an M.2 slot that needs to be enabled in BIOS.

4. Hardware-installation checklist to prevent future failures

- 1 Verify CPU—motherboard socket and BIOS compatibility before purchase/assembly, and update BIOS first if the CPU is newer than the board's release.
- 2 Reseat and visually confirm every power connector (24-pin, CPU 8-pin, GPU PCIe, SATA power) is fully clicked in before first boot.
- 3 Confirm RAM is installed in the motherboard-recommended slot configuration and reads the correct total capacity in BIOS before installing an OS.
- 4 Confirm each storage device appears correctly in BIOS with the right capacity before proceeding to OS installation.
- 5 Perform a breadboard/minimal-config test on any new component combination before full assembly, to isolate faults early.

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Section D — AI-Augmented Learning

Step 1 — Prompt given to the AI tool :-

"Create a step-by-step troubleshooting checklist for a junior IT technician diagnosing a desktop PC that fails to boot immediately after assembly. Cover the motherboard, RAM, storage, and PSU as possible failure points, present it as a clear checklist in the correct priority order a technician would actually follow on-site, and then apply it to a scenario where ABC Solutions is deploying a new computer lab and some newly assembled machines fail to start while others show hardware detection errors — show which checklist step would have caught the reported issue."

Step 2 — Review without AI: issue found and correction

Issue identified: the AI's checklist places "check operating system boot order in BIOS settings" (step 5) before reseating the PSU power connectors (step 6), and — more importantly — the checklist does not mention ESD (electrostatic discharge) precautions at any point before physically touching internal components in steps 2, 3, 4, and 6. Checking a software/BIOS setting like boot order before the physical power connections are confirmed is out of order (a board with a loose power connector may not even reach BIOS, making a boot-order check meaningless), and skipping ESD precautions is an unsafe practice that risks damaging components while "fixing" the reported fault.

3–4 line note:

what changed and why I added an explicit ESD precaution as the first step, since the AI's version went straight into opening the case and touching components without it — a genuine safety gap in an otherwise sound checklist. I also reordered the PSU power-connector check to run before the RAM and storage checks, and moved the BIOS boot-order check to the very end, because a software setting cannot be

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meaningfully diagnosed until the physical power path to the board has already been confirmed. These two fixes make the checklist both safer and logically ordered from "can the board power on at all" through to "software configuration," which better matches how a technician would actually work through a dead machine on-site.